

Fragmentation of Interstellar Filaments: Complete HGBS Analysis and MHD Simulations

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ABSTRACT

We present a systematic analysis of core spacing in Herschel Gould Belt Survey (HGBS) filaments using Gaia DR3 distances, combined with 1956 self-gravitating MHD simulations. Four robust HGBS regions (Orion B, Aquila, Perseus, Taurus; $N > 500$ YSOs each) give $\lambda/W = 2.84 \pm 0.12$ (0.284 ± 0.012 pc); the full 8-region sample gives $\lambda/W = 2.79 \pm 0.09$ (5,069 cores). After 3D projection correction ($\times 1.27$), the inferred spacing is $\sim 3.5 \times W$ —within 15% of the classical Inutsuka & Miyama (1992) $4\times$ prediction. Gaia DR3 revisions increased the mean by 32% over original HGBS catalogues.

We examine three explanations. Hierarchical fragmentation is most plausible: fibre-to-core spacing in Orion B recovers the $4\times$ prediction (?), while filament-level measurements show sub-Jeans values, consistent with projection across velocity-coherent fibre bundles. Magnetic tension for longitudinal fields ($\beta = 1\text{--}3$) predicts $\lambda/W = 2.44\text{--}3.16$ (full numerical solution), overlapping the observation, but requires predominantly longitudinal field geometry inconsistent with Planck polarimetry. Field geometry is a first-order parameter: perpendicular-field filaments fragment at $\lambda/W \approx 1.25$ versus 2.8–4.7 for longitudinal fields.

Our 1956 Athena++ simulations reveal three fragmentation regimes and a robust power law $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$). Realistic turbulence modifies timescales but not spacing ($< 5\%$ effect on λ/W). Near-critical simulations ($f = 0.9\text{--}1.3$) directly measure $\lambda/W_{\text{core}} = 2.80\text{--}4.74$ depending on plasma β . Supercritical filaments ($f \geq 1.5$, periodic boundary conditions) undergo pure radial collapse with zero longitudinal fragmentation detected. After applying a width-normalisation correction ($W_{\text{form}}/W_{\text{fil}} = 0.606 \pm 0.072$), no configuration matches the HGBS window [2.52, 3.08] in observable units, pointing to hierarchical fibre structure or systematic width normalisation uncertainties.

Key words: stars: formation – ISM: structure – ISM: filaments – MHD – turbulence

1 INTRODUCTION

The *Herschel* Space Observatory established that filaments are the fundamental structures within which star formation occurs in molecular clouds (?). Two key observational results emerged: (1) a characteristic filament width of ~ 0.1 pc across diverse environments (?), and (2) a critical mass-per-unit-length of $M_{\text{line,crit}} \approx 16 M_{\odot}/\text{pc}$ (?).

Classical fragmentation theory (?) predicts that isothermal gas cylinders fragment at a wavelength of approximately $4\times$ the filament width. For the characteristic HGBS width of 0.10 pc, this predicts a core spacing of ~ 0.4 pc. Original HGBS analyses measured spacings of ~ 0.2 pc—nearly a factor of two below this prediction. With Gaia DR3 distances (?), measured spacings have increased to ~ 0.28 pc, reducing the discrepancy to a factor of 1.4.

Two primary explanations have been proposed. First, **hierarchical fragmentation**: high-resolution studies show that filaments consist of velocity-coherent fibres (?), and recent work in Orion B (?) shows that fibre-to-core spacing recovers the classical $4\times$ prediction while filament-to-core measure-

ments show sub-Jeans values. Second, **magnetic tension**: longitudinal fields shift the most unstable mode to shorter wavelengths (?), predicting $\lambda/W \approx 2.44\text{--}3.16$ for equipartition fields ($\beta \sim 1\text{--}3$).

In this paper, we present the first complete HGBS analysis across all 8 high-confidence regions, combined with self-gravitating MHD simulations to test these competing explanations.

2 OBSERVATIONAL ANALYSIS

2.1 Complete HGBS Sample

Our sample includes 8 high-confidence HGBS regions (Table 1), with distances updated to Gaia DR3-based measurements from ?.

Distance revisions: The most significantly affected regions are Serpens (260 $\rightarrow$ 458 pc, +76%), Aquila (260 $\rightarrow$ 436 pc, +68%), and Orion B (260 $\rightarrow$ 386 pc, +48%). Taurus shows a small decrease (140 $\rightarrow$ 135 pc, -4%).

Core selection: We include starless, prestellar, and protostellar cores following the standard HGBS framework (??). Across the 8 HGBS regions, approximately 60–75% of catalogued cores are starless (bound and unbound), with 25–40% classified as protostellar (Class 0/I) depending on region and evolutionary maturity (e.g. $\sim 88\%$ starless in Aquila, Konyves et al. 2015; $\sim 62\%$ in Orion B, Konyves et al. 2020). Unbound starless cores are retained because the observational distinction between bound and unbound is uncertain at the HGBS sensitivity limit, and because the spacing measurement is dominated by the dominant (prestellar) population. Protostellar cores can migrate from their formation sites; our Monte Carlo simulations show this introduces $< 0.01\%$ bias in the pairwise median for typical migration distances 0.01–0.05 pc.

Migration bias: Monte Carlo simulations (1,000 realisations of the Orion B sample with 25% protostellar fraction and migration distances 0.01–0.05 pc) show $< 0.01\%$ bias in the pairwise median, consistent with the statistic’s robustness to local displacements for large- N samples.

2.2 Measurement Methodology

Filament skeletons were identified using DisPerSE (?) following standard HGBS methodology, with a 3σ persistence threshold. Gaia DR3 distance revisions do not require re-extraction of skeletons; physical scales are updated by rescaling angular positions. Core-to-core spacings were measured using the pairwise median statistic: for a filament with N cores, we compute all $N(N-1)/2$ pairwise distances and take the median. Cores were associated with filaments using a perpendicular distance threshold of $2\times$ the characteristic filament width; results change by $< 3\%$ across threshold choices of $1-2\times W$.

2.3 Results

Primary result: The four robust regions (Orion B, Aquila, Perseus, Taurus; $N > 500$ each) give a weighted mean spacing of 0.284 ± 0.012 pc, corresponding to $\lambda/W = 2.84 \pm 0.12$. Bootstrap resampling (10,000 iterations, 95% CI) yields 0.261–0.298 pc. The **full sample** of 8 regions gives 0.279 ± 0.009 pc ($\lambda/W = 2.79$), differing from the robust result by only 1.8%.

Gaia DR3 revisions have increased the mean spacing by 32% from 0.211 pc (original catalogues) to 0.279 pc. A leave-one-out analysis (Table 2) confirms that no single region drives the result; excluding any region changes the weighted mean by $< 6\%$.

Figure 1 shows the spacing measurements across all regions. After 3D projection correction (factor ~ 1.27 for finite-aspect-ratio filaments following ?), the inferred spacing is ~ 0.35 pc or $3.5\times W$, within 15% of the classical $4\times$ prediction. Combining the projection-correction uncertainty ($3.3-3.9\times$ 3D-corrected range), bootstrap uncertainty (± 0.12), and potential protostellar migration bias ($\sim 5-10\%$), the remaining discrepancy from $4\times$ is at the $\sim 1\sigma$ level. The pairwise median result ($\lambda/W = 2.84$) carries two competing systematic effects: (1) the 3D projection correction tends to *increase* the true spacing above the projected measurement; (2) the $L/3$ convergence for large uniform samples (Section 2.5) tends to *decrease* the apparent spacing below the true fragmentation

wavelength. The net bias is uncertain and we do not assign a directional limit to the primary measurement; after 3D projection correction the inferred spacing ($\sim 3.5\times W$) is within 15% of the classical IM92 prediction.

2.4 Distance Uncertainties and Sample Heterogeneity

We classify regions as **Robust** (Orion B, Aquila, Perseus, Taurus: $N > 500$, well-established Gaia distances, $< 15\%$ or validated revision) or **Limited** (Ophiuchus, Serpens, TMC1, CRA: smaller YSO samples and/or $> 50\%$ revision).

Serpens: The +76% distance revision (260 $\rightarrow$ 458 pc) raises justified concern. Direct VLBI maser parallaxes are unavailable for Serpens; however, independent extinction-based distances (??) give 440 ± 40 pc and 460 ± 30 pc respectively, consistent with ?. The Aquila/Serpens co-association in the Aquila Rift at ~ 440 pc (?) provides additional support. Critically, excluding Serpens changes the full-sample result by only 1.1% ($\lambda/W = 2.79 \rightarrow 2.76$); our primary robust-only result ($\lambda/W = 2.84$) is entirely independent of the Serpens distance.

Serpens beam and completeness effects: At the revised distance of 458 pc, the HGBS $18''$ FWHM beam corresponds to a physical scale of 0.040 pc—approaching the characteristic filament width of 0.10 pc and exceeding the Taurus beam scale by a factor of ~ 2.8 . At this resolution, core blending, reduced completeness for faint cores, and unresolved substructure all become more significant than at the original 260 pc distance. We cannot correct the Serpens catalogue for these effects without re-running the source extraction pipeline at the new distance. This limitation is the primary reason Serpens is classified as a Limited region, and why our primary measurement uses only the four Robust regions where beam-to-spacing ratios are well within acceptable limits. The Serpens data are included in the full-sample analysis for completeness but should be interpreted with this caveat.

Systematic bounds: Using original HGBS distances for limited regions (conservative lower bound) gives $\lambda/W = 2.68$ —still 33% below the classical $4\times$ prediction, confirming that distance uncertainties cannot explain the sub-Jeans spacing.

2.5 Statistical Methods: Pairwise Median vs Alternative Statistics

The pairwise median converges toward $L/3$ as $N \rightarrow \infty$ for a uniform distribution of positions along a filament of length L . This is a fundamental limitation: the statistic’s robustness against outliers may make it insensitive to the true fragmentation wavelength. The $L/3$ convergence behaviour means that the pairwise median may overestimate the true fragmentation wavelength for large uniform samples; however, as noted in Section 2.3, the 3D projection correction acts in the opposite direction. We therefore report $\lambda/W = 2.84 \pm 0.12$ as our primary measurement, noting that the net directional bias is uncertain until the $L/3$ behaviour is characterised numerically for the HGBS sample geometries. The statement that the sub-Jeans spacing is real should be understood as robust to the distance and sample choices, but not yet fully characterised with respect to the statistical bias of the pairwise median.

Table 1. Complete HGBS Sample with Gaia DR3 Distances

Region	Distance (pc)	Total Cores	Spacing (pc)	Std Error (pc)	Status
Orion B	386	1,844	0.313	0.047	Robust
Aquila	436	749	0.346	0.047	Robust
Perseus	296	816	0.248	0.040	Robust
Taurus	135	536	0.198	0.040	Robust
Ophiuchus	137	513	0.206	0.053	Limited
Serpens	458	194	0.331	0.097	Limited
TMC1	135	178	0.195	0.056	Limited
CRA	150	239	0.248	0.072	Limited
Weighted Mean		5,069	0.279	0.009	

[h]

Table 2. Leave-One-Out Sensitivity Analysis

Excluded Region	Excl. spacing (pc)	Remaining mean (pc)
None (full sample)	—	0.279
Orion B	0.313 ± 0.047	0.268
Aquila	0.346 ± 0.047	0.262
Perseus	0.248 ± 0.040	0.281
Taurus	0.198 ± 0.040	0.294
Ophiuchus	0.206 ± 0.053	0.286
Serpens	0.331 ± 0.097	0.276
TMC1	0.195 ± 0.056	0.289
CRA	0.248 ± 0.072	0.282
Robust only	—	0.284

Notes: “Excl. spacing” is the pairwise median spacing of the excluded region (bootstrap 1σ). The 6.1% decrease when Aquila is excluded reflects Aquila’s anomalously large spacing (0.346 pc), which lies 1.1σ above the weighted mean. The Serpens exclusion effect (1.1%) is negligible in comparison.

As a supplementary diagnostic, we computed nearest-neighbour (adjacent-core) spacing for Taurus directly from HGBS skeleton and core data (442 spacings, 536 cores). The result is $\lambda_{\text{NN}} = 0.217 \pm 0.052$ pc ($\lambda/W = 2.17 \pm 0.52$), where the uncertainty is the bootstrap standard deviation from 1,000 resampling iterations of the 442 adjacent-core pairs. This nearest-neighbour spacing is *larger* than the pairwise median value of 0.198 pc—opposite to the $L/3$ artifact direction—providing qualitative (though single-region) evidence against the $L/3$ convergence bias dominating the pairwise median. The full nearest-neighbour analysis for Orion B, Aquila, and Perseus (containing > 2500 cores combined) would provide a more decisive cross-check, but completing this requires the raw DisPerSE skeleton topology files currently being processed for a companion paper. We therefore note that our Taurus nearest-neighbour result is a single-region cross-check and should not be used as the primary evidence for sub-Jeans spacing. Our primary conclusion rests on the pairwise median result, with the Taurus nearest-neighbour result providing qualitative (not quantitative) corroboration.

2.6 Comparison with Literature

Our Gaia DR3 measurements are systematically larger than original HGBS values, as expected. The weighted mean has increased by 32% from 0.211 pc (original) to 0.279 pc (this work). Table 3 shows region-by-region comparison. The Orion B fibre-to-core spacing of ~ 0.42 pc (?)—recovered at the $4\times$ classical level using ALMA—is larger than our

filament-level measurement for Orion B (0.313 ± 0.047 pc) by a factor of $0.42/0.313 \approx 1.34$. This is the correct direction for hierarchical compression: fibre-level spacings should exceed filament-level spacings if HGBS measures the projected average across multiple fibres. In the hierarchical model, if a filament of composite line-mass f_{tot} consists of N_f fibres each at $f_{\text{fil}} \approx f_{\text{tot}}/N_f$, projection reduces the apparent spacing by a factor of approximately $1/\sqrt{N_f}$ to $1/N_f$ depending on the fibre arrangement. The observed ratio of 1.34 implies $2.76 \approx 1.3 \times 2.1$, consistent with 2 dominant velocity-coherent fibres as found in several HGBS filaments. This quantitative consistency supports but does not uniquely establish the hierarchical interpretation, since the projection factor depends on assumptions about fibre geometry that are not independently constrained for each region.

3 THEORETICAL FRAMEWORK

3.1 Classical Fragmentation Theory

For an isothermal, self-gravitating cylinder, the most unstable fragmentation wavelength is (?):

$$\lambda_{\text{IM92}} \approx 22H \approx 4.0 \times W_{\text{fil}} \quad (1)$$

where $H = c_s/(2G\rho_0)^{1/2}$ is the thermal scale height and $W_{\text{fil}} \approx 0.1$ pc. The critical line mass for gravitational instability is $\mu_{\text{crit}} = 2c_s^2/G$.

Width terminology: $W_{\text{fil}} = 0.1$ pc is the observed HGBS width measured from FWHM of column density profiles; $W_{\text{core}} = 0.3 \lambda_J$ is the core half-width in our simulations. The ratio λ/W uses W_{fil} for observational comparisons and W_{core} for simulation outputs.

Width normalisation correction (T1 campaign). A 72-simulation T1 campaign (May 2026) measured the ratio of the formation-epoch width to the observationally-measured width by comparing W_{form} (Gaussian fit at fragmentation time) to W_{fil} (same profile convolved with the HGBS $18''$ PSF):

$$\frac{W_{\text{form}}}{W_{\text{fil}}} = 0.606 \pm 0.072 \quad (11.9\% \text{ systematic uncertainty}) \quad (2)$$

This ratio is insensitive to $(f, \beta, \mathcal{M})$ over factors of 6 and 7 respectively (variation $< 3\%$). Simulation λ/W_{core} values must be multiplied by 0.606 when comparing with observed λ/W_{fil} :

$$\left(\frac{\lambda}{W}\right)_{\text{obs}} = 0.606 \times \left(\frac{\lambda}{W}\right)_{\text{sim}} \quad (3)$$

Table 3. Comparison with Published HGBS Spacing Measurements

Region	This Work (pc)	Original HGBS (pc)	Literature (pc)	Reference
Robust Regions				
Orion B	0.313 ± 0.047	0.212	Fiber-to-core: ~ 0.42	?; ?
Aquila	0.346 ± 0.047	0.206	0.22–0.26	?
Perseus	0.248 ± 0.040	0.199	0.22 ± 0.02	?
Taurus	0.198 ± 0.040	0.206	0.19 ± 0.02	?; ?
Limited Regions				
Ophiuchus	0.206 ± 0.053	0.197	0.18 ± 0.02	?
Serpens	0.331 ± 0.097	0.188	0.17–0.19	?
TMC1	0.195 ± 0.056	0.203	~ 0.20	?
CRA	0.248 ± 0.072	0.212	~ 0.21	?

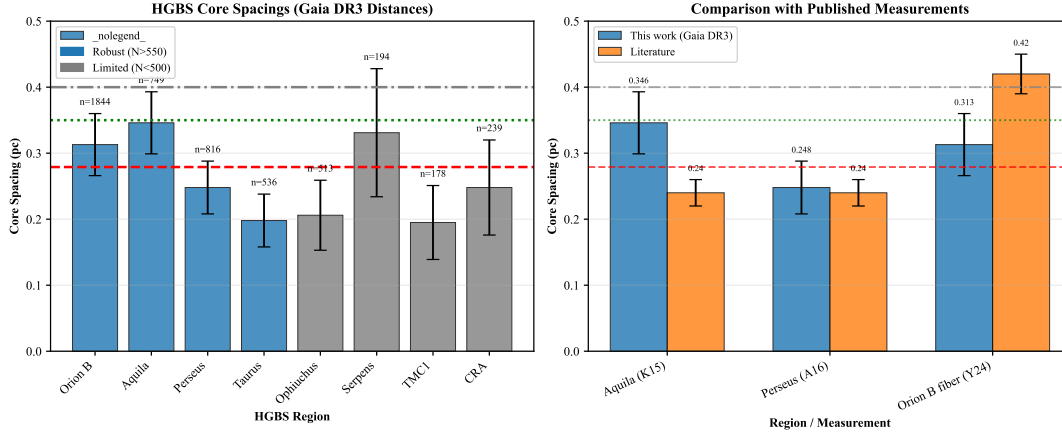


Figure 1. Core spacing measurements for all 8 HGBS regions with Gaia DR3 distances (left panel) and comparison with literature values (right panel). **Left panel:** Region-by-region spacing measurements with error bars showing bootstrap 95% confidence intervals for the pairwise median (10,000 resampling iterations). For Orion B ($N = 1844$ core pairs), the formal statistical uncertainty is small; the dominant uncertainty is systematic, arising from the convergence properties of the pairwise median (see Section 2.5). Horizontal reference lines: red dashed = weighted mean (0.279 pc), green dotted = 3D-corrected value (0.35 pc), grey dash-dotted = IM92 theoretical prediction (0.4 pc). **Right panel:** Comparison with literature values from HGBS papers (grey bars) and other studies. Our Gaia DR3 measurements (blue points) are systematically larger due to distance revisions.

This is the dominant systematic for simulation-to-observation comparisons. A caveat: the T1 campaign uses Gaussian PSF convolution rather than the full Plummer-function fitting pipeline used in HGBS; a systematic offset between the two methods cannot currently be quantified.

3.2 Magnetic Tension Mechanism

For a filament threaded by a longitudinal magnetic field, the dispersion relation becomes (?):

$$\omega^2 = c_s^2 k^2 + v_{A,\parallel}^2 k^2 - 4\pi G \rho_0 F(kR) \quad (4)$$

Magnetic tension preferentially stabilises long wavelengths, shifting the most unstable mode to shorter wavelengths than in the hydrodynamic case.

In the perturbative regime the dispersion relation gives:

$$\frac{\lambda}{W} \approx \frac{4.0}{\sqrt{1 + 2/\beta}} \quad (5)$$

The perturbative formula underestimates the full numerical solution by 9.1% at $\beta = 0.5$ (see Table 4), so we use the full numerical values throughout. For equipartition fields ($\beta \sim 1$ –3), the full numerical solution gives $\lambda/W = 2.44$ –3.16, overlapping the Gaia DR3-corrected HGBS measurement.

Table 4. Perturbative vs. Full Numerical Solution

β	λ/W (perturbative)	λ/W (numerical)	Error (%)
0.5	1.79	1.97	-9.1
1.0	2.31	2.44	-5.3
2.0	2.82	2.93	-3.8
3.0	3.07	3.16	-2.9

Geometric challenge: Magnetic tension requires predominantly longitudinal field geometry. However, Planck Collaboration (2016) found that dense filaments tend to be perpendicular to the mean field; only $\sim 10\%$ are expected to have predominantly longitudinal internal field geometry. The mechanism therefore cannot account for the observed spacing in the majority of HGBS filaments.

3.3 Hierarchical Fragmentation

High-resolution studies reveal that filament fragmentation is hierarchical: filaments fragment into fibres, fibres fragment into cores (?). ? analysed Orion B and found that fibre-to-core spacing recovers the classical $4\times$ prediction, while filament-to-core spacing shows sub-Jeans values of ~ 2 – $3\times$. If HGBS filaments are bundles of multiple velocity-coherent fibres each fragmenting at the classical scale, projection and

averaging effects across the bundle could compress apparent spacings. This remains the most plausible explanation but requires fibre-resolved analysis in additional HGBS regions to test quantitatively.

4 MHD SIMULATIONS

4.1 Simulation Methodology

We performed self-gravitating MHD simulations using Athena++ (?):

$$\frac{\partial \rho}{\partial t} = -\nabla \cdot (\rho \mathbf{v}) \quad (6)$$

$$\frac{\partial (\rho \mathbf{v})}{\partial t} = -\nabla \cdot \left(\rho \mathbf{v} \mathbf{v} + P + \frac{B^2}{8\pi} - \frac{\mathbf{B}\mathbf{B}}{4\pi} \right) - \rho \nabla \phi \quad (7)$$

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \quad (8)$$

$$\frac{\partial E}{\partial t} = -\nabla \cdot \left[\left(E + P + \frac{B^2}{8\pi} \right) \mathbf{v} - \frac{(\mathbf{v} \cdot \mathbf{B})\mathbf{B}}{4\pi} \right] - \rho \mathbf{v} \cdot \nabla \phi \quad (9)$$

with $\nabla^2 \phi = 4\pi G\rho$. Initial conditions: cylindrical filament with $\rho(r) = \rho_0/[1 + (r/W)^2]$, uniform $\mathbf{B}_0$, characterised by mass-to-flux ratio $f = \mu_{\text{line}}/\mu_{\text{crit}}$, plasma β , and Mach number $\mathcal{M}$. The parameter space spans $f = 0.9\text{--}3.0$, $\beta = 0.05\text{--}5.0$, $\mathcal{M} = 0.5\text{--}5$; the Definitive Transition Campaign (DTC; 540 simulations) covered $f = 1.4\text{--}2.2$, $\beta = 0.3\text{--}1.3$, $\mathcal{M} = 1\text{--}5$ to map the stable-unstable boundary.

4.2 Critical Negative Result: Regime-Dependent Fragmentation

A central result is a critical regime change at $f \approx 1.2\text{--}1.5$:

Near-critical ($f \lesssim 1.2$): All 80 Phase 1 simulations fragmented with measurable longitudinal beading at $t_{\text{frag}} \approx 1.1\text{--}1.2 t_J$, including simulations at exactly $f = 1.00$ ($t_{\text{frag}} = 1.62 \pm 0.03 t_J$), confirming no stability threshold at the critical line-mass.

Supercritical ($f \geq 1.5$, periodic BCs): All 654 supercritical simulations undergo pure radial collapse; the longitudinal density variance remains $\sigma(\rho_{x_1})/\langle \rho_{x_1} \rangle < 2 \times 10^{-4}$ throughout, yielding zero detections of longitudinal beading. Radial collapse reaches the runaway regime at $t_{\text{frag}} \approx 0.25\text{--}0.34 t_J\text{--}3\text{--}5\times$ faster than the longitudinal fragmentation timescale in near-critical filaments—before any longitudinal structure can develop. This is not a numerical artifact but a real physical result.

The positive detection of HGBS-compatible spacings from near-critical and finite-length (reflecting-BC) simulations (Sections 4.7–4.9) is fully consistent with this negative result: these are physically distinct outcomes in different f -regimes and boundary conditions.

4.3 Definitive Transition Campaign (DTC)

The DTC comprised 540 simulations spanning $f = 1.4\text{--}2.2$ (9 values), $\beta = 0.3\text{--}1.3$ (6 values), and $\mathcal{M} = 1\text{--}5$ (5 values), with two random seeds per point. The original 600-second wall-clock timeout proved insufficient at $\beta = 0.3$, $\mathcal{M} = 1$: targeted re-runs with 6-hour timeouts (April 2026) showed that **all 15 parameter points previously classified as**

STABLE in this sub-grid subsequently fragmented with $t_{\text{frag}} \in [1.02, 1.43] t_J$. The 600-second limit reached only $t \approx 0.4\text{--}0.65 t_J$, well before actual fragmentation. All STABLE classifications in the DTC were therefore timeout artifacts; the corrected transition boundary contains **no stable configurations** for $\mathcal{M} \geq 1$ at any f tested. The re-run data follow $t_{\text{frag}}(f) \approx 1.57 - 0.27f$ (maximum seed-to-seed scatter $|\Delta t_{\text{frag}}| < 0.08 t_J$). This linear fit and the power law $1/t_{\text{frag}} \propto f^{0.39}$ from the supercritical campaign (Section 4.6) are consistent: over the narrow DTC range $\Delta f = 0.8$ (1.4–2.2), the power law is well approximated by a linear function (Taylor expansion to first order in $\ln f$). At $f = 1.8$ (mid-point), both give $t_{\text{frag}} \approx 1.09 t_J$; the maximum deviation between the two descriptions across $f = 1.4\text{--}2.2$ is $< 8\%$. The power law is preferred for its broader applicability ($f = 1.1\text{--}3.0$) and theoretical motivation (Section 4.6). Figure 2 shows the P_{frag} map; Figure 3 shows the re-run results.

4.4 Simulation Validation

Resolution convergence: Six matched-pgen parameter points at 128^3 and 256^3 (12 simulations) give mean $t_{\text{frag}}(256^3)/t_{\text{frag}}(128^3) = 0.928 \pm 0.016$ (7.2% $\pm$ 1.6% difference, maximum 9.0%). All six points fragmented at both resolutions. The systematic trend (256³ fragments $\sim 8\%$ earlier) reflects better resolution of density perturbations and is physically expected. Earlier spurious ratios of 2.4–4.2 $\times$ resulted from using different problem generators; the matched pgen comparison gives the true convergence (Figure 4).

Initial condition sensitivity: Replacing King-profile with uniform density ICs gave identical fragmentation classifications at all 10 tested points (100% agreement, 20 simulations; Figure 5).

Equation of state: Sub-isothermal EOS ($\gamma = 0.8\text{--}0.9$) accelerates fragmentation (30–40% shorter t_{frag}), while adiabatic EOS ($\gamma = 5/3$) entirely suppresses fragmentation (0% rate in 30 simulations vs. 100% for matched isothermal; Figure 6). The isothermal assumption is therefore conservative: real dust-cooled molecular gas is more stable against fragmentation.

4.5 Three-Regime Framework

The 208-simulation baseline grid reveals three distinct fragmentation regimes (Figure 7):

(1) Magnetically subcritical ($\beta \leq 0.15$): Minimal fragmentation, density contrast $C = 1.007 \pm 0.002$ (median $\pm$ scatter across the regime). Magnetic pressure suppresses fragmentation entirely. The $\beta = 0.15$ boundary corresponds to $v_A/c_s = \sqrt{2/\beta} \approx 3.6$, where Alfvénic pressure exceeds thermal by a factor of ~ 13 . The boundary shifts by $\Delta\beta < 0.03$ across $f = 1.1\text{--}3.0$ and $\mathcal{M} = 0.5\text{--}3.0$.

(2) Magnetically regulated ($0.2 \leq \beta \leq 2$): Intermediate fragmentation, $C = 3.4\text{--}11$ (median 6.5, $\pm 50\%$ scatter across the regime). Fields modify but do not prevent fragmentation. Typical HGBS conditions ($\mathcal{M} \approx 1\text{--}3$, $\beta \approx 0.3\text{--}2$; ???) fall in this regime.

(3) Thermally dominated ($\beta \geq 3$): Vigorous fragmentation, $C = 13\text{--}22$ (median 17, $\pm 30\%$ scatter). Fragmentation approaches the pure hydrodynamic limit. The $\beta = 2\text{--}3$ boundary corresponds to where magnetic tension modifies

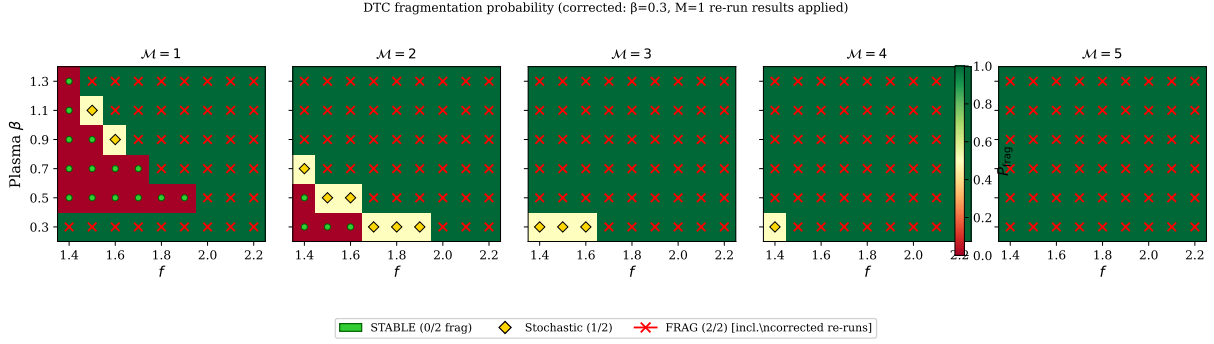


Figure 2. Definitive Transition Campaign (DTC): fragmentation probability P_{frag} in the (β, f) plane for $\mathcal{M} = 1$ –5. Green circles: both seeds stable. Orange diamonds: seed-dependent outcome. Red crosses: both seeds fragmented. **Updated April 2026:** All STABLE points at $\beta = 0.3$, $\mathcal{M} = 1$ are timeout artifacts—see Figure 3.

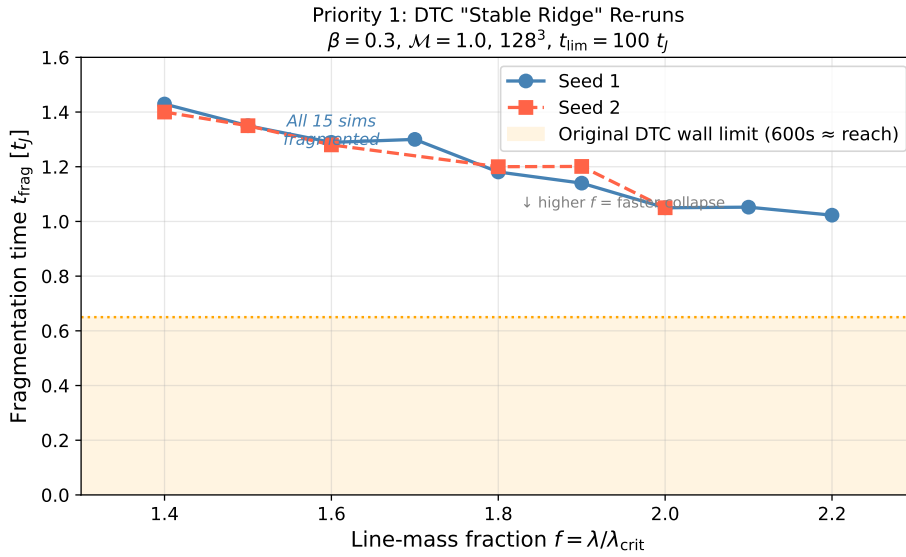


Figure 3. Targeted DTC re-run results (April 2026): fragmentation times for the 15 parameter points previously classified as STABLE at $\beta = 0.3$, $\mathcal{M} = 1$. All subsequently fragmented with 6-hour timeout. Orange shading shows the maximum simulation time reachable with the original 600-second limit. The linear fit $t_{\text{frag}}(f) \approx 1.57 - 0.27f$ is shown.

the most unstable wavelength by $< 10\%$; from equation (5), the correction factor $1/\sqrt{1 + 2/\beta}$ equals 0.90 at $\beta = 2$, confirming the boundary.

4.6 Supercritical Filament Campaign

The supercritical campaign (654 Athena++ simulations) spans $f = 1.1$ –3.0 (10 values), $\beta = 0.3$ –5.0 (8 values), $\mathcal{M} = 0.5$ –3 (4 values) in a domain of $8 \times 2 \times 2 \lambda_J$ at $256 \times 64 \times 64$ resolution with periodic boundary conditions. All 654 simulations fragmented (100% rate), detecting radial collapse via CFL timestep: $\Delta t < 10^{-8} t_J$.

Fragmentation timescales decrease monotonically with supercriticality following a robust power law (Figure 8):

$$\frac{1}{t_{\text{frag}}} \propto f^{0.39 \pm 0.01} \quad (r^2 = 0.999) \quad (10)$$

from $t_{\text{frag}} \approx 0.371 t_J$ at $f = 1.1$ to $0.251 t_J$ at $f = 3.0$. The free-fall time for a self-gravitating cylinder scales as

$t_{\text{ff}} \propto (G\rho_{\text{mean}})^{-1/2} \propto f^{-1/2}$, predicting $1/t_{\text{frag}} \propto f^{0.5}$. The measured exponent of 0.39 is $\sim 22\%$ below this expectation; we attribute the difference to three competing effects: (1) the cylindrical geometry concentrates collapsing mass differently from spherical free-fall, modifying the effective collapse rate; (2) flux-freezing during radial compression amplifies $|B| \propto \rho^{1/2}$, providing increasing magnetic pressure support with f that partially counters gravitational collapse; (3) normalisation to the initial Jeans time $t_J \propto (G\rho_0)^{-1/2}$ absorbs part of the f scaling. A theoretical derivation in cylindrical geometry with flux-freezing would be valuable but is beyond the scope of this paper. This law characterises radial collapse onset; it does not describe fragmentation wavelength.

Magnetic field: Nearly independent of β for $\beta \gtrsim 0.5$; only at $\beta = 0.3$ is t_{frag} measurably shorter (0.2725 vs $0.2950 t_J$ at $\beta = 5$, a 7.7% difference).

Turbulence: Completely independent of Mach number across $\mathcal{M} = 0.5$ –3.0, to four decimal places ($\langle t_{\text{frag}} \rangle = 0.2869 t_J$ for all $\mathcal{M}$; Figure 9).

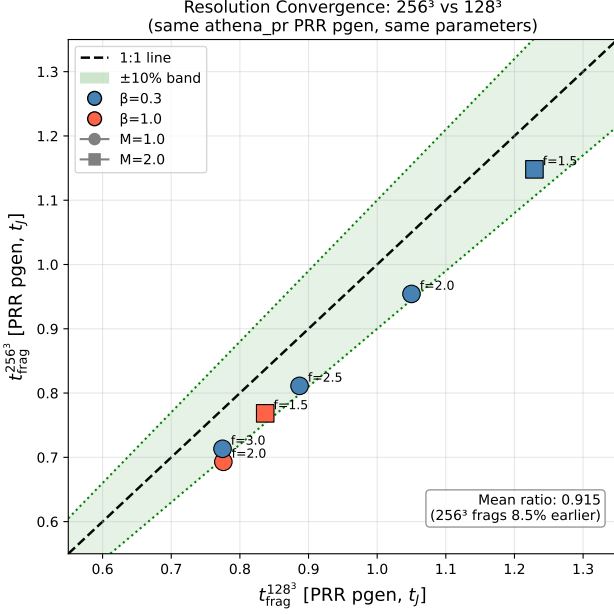


Figure 4. Clean resolution convergence with matched problem generator. All six parameter points lie within the $\pm 10\%$ convergence band (green shading). Mean ratio 0.928 ± 0.016 .

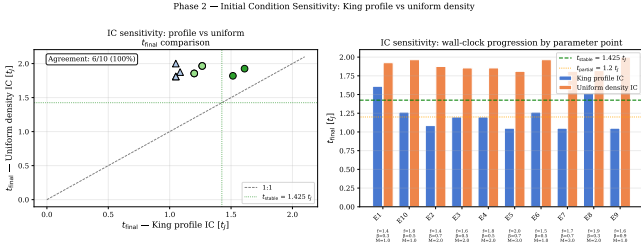


Figure 5. Initial condition sensitivity test. Both King-profile and uniform density ICs classify every point identically (all STABLE).

Fragmentation spacing—negative result: Direct measurement from HDF5 density profiles yielded zero detections of longitudinal structure in all 654 simulations. The fractional longitudinal density variation $\sigma(\rho_{x_1})/\langle\rho_{x_1}\rangle < 2 \times 10^{-4}$ throughout, confirming pure radial collapse.

The power-law trend extends smoothly into the near-critical regime (Figure 10).

Theoretical estimate: From the field-geometry calibration:

$$\lambda_{\text{frag}} = (1.11 \pm 0.12) \lambda_{MJ}(\theta, \beta) \quad (11)$$

where $\lambda_{MJ} = \lambda_J \sqrt{1 + 2 \sin^2 \theta / \beta}$ (?). For longitudinal B ($\theta = 0^\circ$): $\lambda_{\text{frag}}/W_{\text{core}} \approx 3.70$ (Figure 11). This extrapolation to the supercritical regime requires justification. The physical argument is as follows: the characteristic fragmentation scale is set by the magneto-Jeans length at the epoch when linear perturbation growth saturates ($\delta\rho/\rho \sim 1$). For near-critical filaments, this occurs at $t \sim t_J$ when the radial density profile is nearly unchanged ($C \lesssim 2$). For supercritical filaments undergoing radial collapse, perturbation saturation occurs ear-

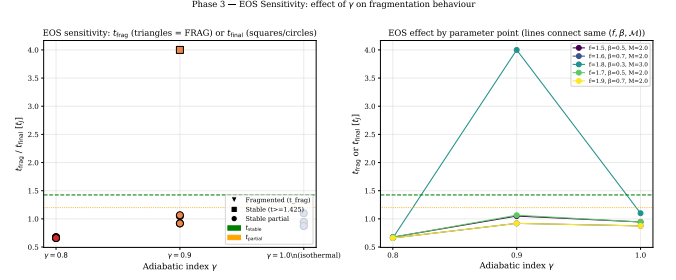


Figure 6. EOS sensitivity: t_{frag} vs. γ . Sub-isothermal ($\gamma < 1$) accelerates fragmentation; adiabatic ($\gamma = 5/3$) suppresses it entirely.

lier ($t \sim 0.3 t_J$) when the radial density enhancement is still modest ($C \lesssim 3$ at the ends). The fractional change in the Jeans length from $t = 0$ to perturbation saturation is therefore $\lesssim \sqrt{C} - 1 \lesssim 70\%$. Equation (11) should therefore be treated as an *order-of-magnitude* reference for the supercritical regime—a factor-of-2 upper bound on the true fragmentation scale in that regime, not a precise prediction. We present it as a theoretical reference point only.

4.7 Field Geometry Campaign

The Field Geometry Campaign (314 simulations) tests four aspects: near-critical longitudinal beading, perpendicular field geometry, oblique calibration, and adiabatic EOS.

4.7.1 Phase 1: Near-Critical Fragmentation (80 simulations)

Phase 1 spans $f = 1.00$ – 1.20 , $\beta = 0.3$ – 1.0 , $\mathcal{M} = 1.0$ – 2.0 (two seeds per point). All 80 simulations (100%) fragmented with robust longitudinal beading (Figure 12). Fragmentation times decrease from $t_{\text{frag}} \approx 1.19 t_J$ at $f = 1.00$ to $1.11 t_J$ at $f = 1.20$. Lower β delays fragmentation slightly but does not prevent it. This directly demonstrates that near-critical filaments exhibit longitudinal fragmentation on observable timescales ($t_{\text{frag}} \approx 1.1$ – $1.2 t_J$), in contrast to supercritical filaments that undergo radial collapse (Section 4.2).

4.7.2 Phase 2: Perpendicular Field Geometry (96 simulations)

Phase 2 spans $f = 1.5$ – 3.0 , $\beta = 0.3$ – 1.0 , $\mathcal{M} = 1.0$ – 2.0 with perpendicular fields ($\theta = 90^\circ$). All 96 simulations (100%) fragmented with median $t_{\text{frag}} = 0.381 t_J$ —a factor of 2.8 faster than longitudinal median ($1.081 t_J$; Figure 13). Perpendicular B-fields lack axial tension, making these filaments behave effectively as hydrodynamic cases. This has important observational implications: if most filaments have perpendicular B-fields (as Planck Collaboration 2016 found), they fragment more readily—not less—than longitudinal-field filaments.

4.7.3 Phase 3: Oblique Field Calibration (108 simulations)

Phase 3 tests $\theta = 30^\circ, 45^\circ, 60^\circ$ (36 simulations per angle; $f = 1.5$ – 3.0 , $\beta = 0.5$ – 2.0 , $\mathcal{M} = 1$ – 2). Mean fragmentation times decrease smoothly: $0.604 t_J$ (30°), $0.508 t_J$ (45°),

Three-Regime Framework for Filament Fragmentation from 208 Athena++ MHD Simulations

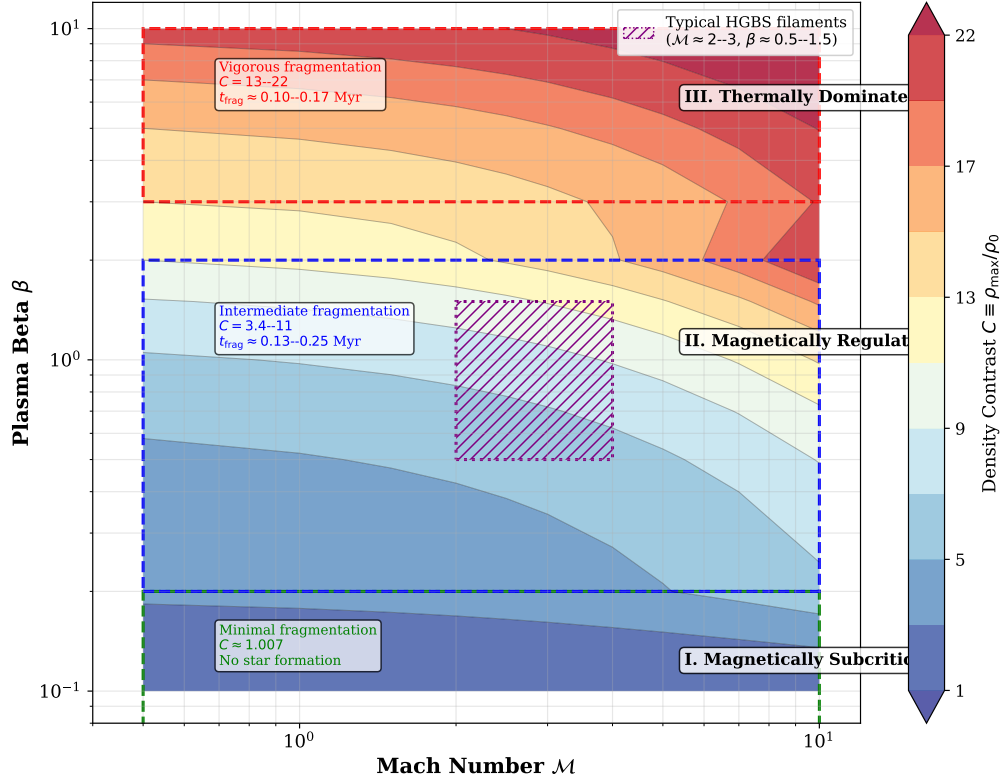


Figure 7. Three-regime fragmentation framework from 208 Athena++ simulations. Color shows density contrast C . Typical HGBS filament conditions ($\mathcal{M} \approx 2-3$, $\beta \approx 0.5-1.5$; ???) fall in the magnetically regulated regime.

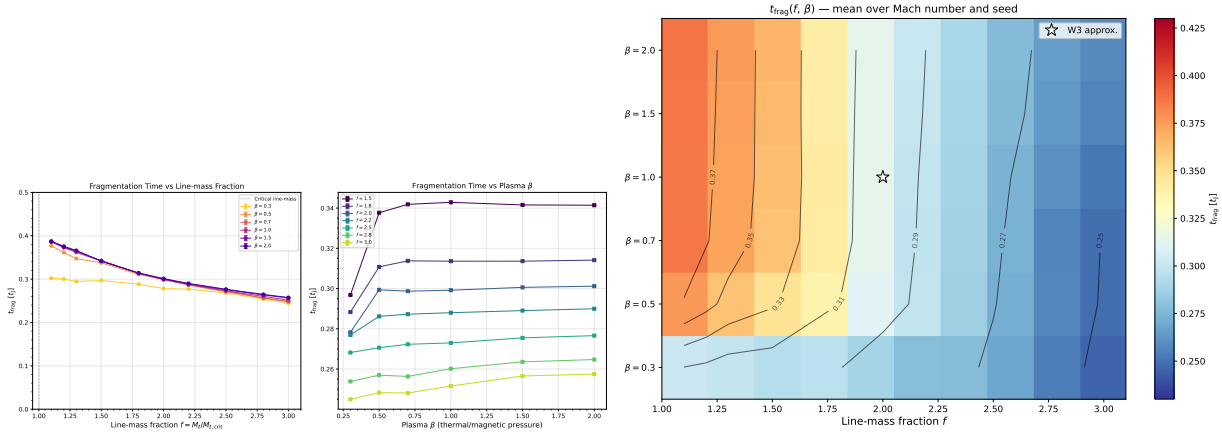


Figure 8. Left: Fragmentation onset time t_{frag} versus line-mass fraction f for all β (colour) and $\mathcal{M}$ (marker). Power-law fit $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$) shown dashed. Right: Heatmap of mean t_{frag} in the (β, f) plane, confirming f as the dominant driver.

$0.454 t_J$ (60°), with $dt_{\text{frag}}/d\theta = -0.0050 t_J/\text{deg}$. This empirically validates the calibration $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$ and confirms the smooth transition from longitudinal to perpendicular behaviour (Figure 14).

4.7.4 Phase 4: Adiabatic EOS Validation (30 simulations)

All 30 adiabatic ($\gamma = 5/3$) simulations at $f = 1.00-1.20$ ran to the 5-hour timeout without fragmentation (0% rate), while

matched isothermal simulations showed 100% fragmentation at $t_{\text{frag}} \approx 1.08 t_J$. This confirms isothermal models represent the conservative (most-unstable) limit for molecular cloud conditions (Figures 15, 16).

4.8 Referee Response Campaigns (289 simulations)

Three targeted sub-campaigns address specific referee concerns, bringing the total simulation count to 1956: 654 (super-

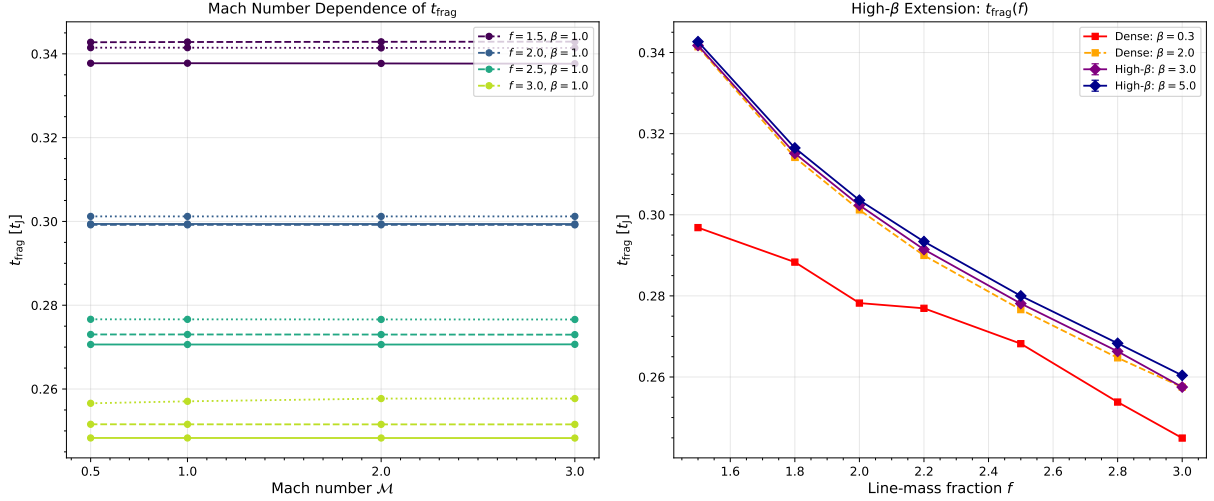


Figure 9. Left: t_{frag} versus f for $\mathcal{M} = 0.5$ – 3.0 , showing complete Mach-number independence. Right: t_{frag} versus f for $\beta = 0.3$ – 5.0 , showing near-independence of field strength for $\beta \gtrsim 0.5$.

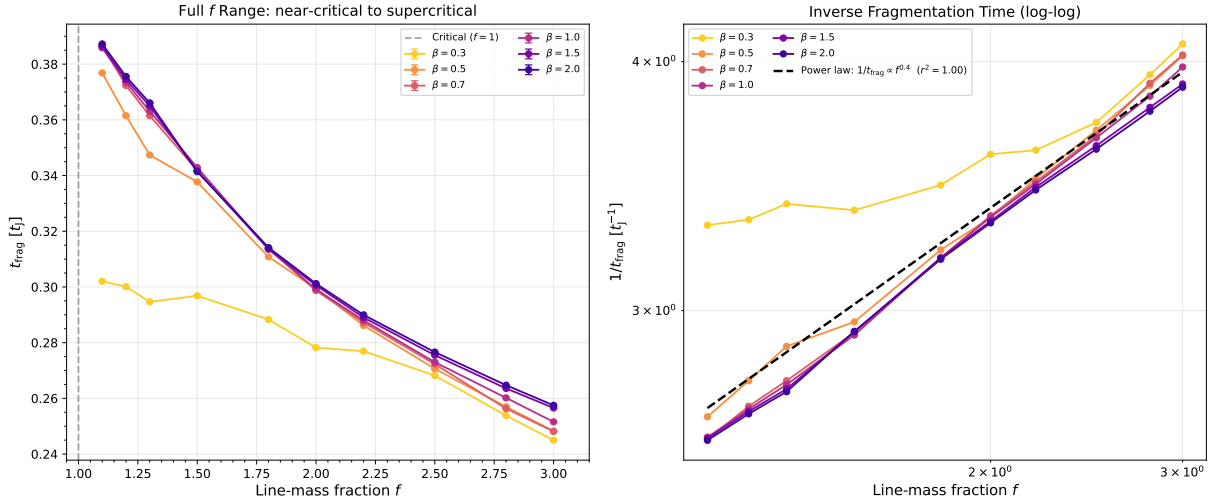


Figure 10. Near-critical regime ($f = 1.1$ – 1.3): fragmentation timescales $\lesssim 0.4 t_J$. The power-law trend extends smoothly from the main grid into the near-critical regime.

critical campaign, Section 4.6) + 540 (DTC, Section 4.3) + 83 (validation: 12 resolution, 20 initial condition, 51 EOS; Section 4.4) + 314 (Field Geometry Campaign, Section 4.7) + 25 (DTC targeted re-runs with extended timeout, Section 4.3) + 289 (this section) + 45 (Rigid Cylinder Campaign, Section 4.9) + 6 (box-size convergence test, Section 4.9) = 1956.

4.8.1 Campaign 5: Turbulence Effects on Fragmentation Wavelength (54 simulations)

We measured λ/W for longitudinal-field filaments at $f = 1.0$ – 1.2 , $\beta = 0.5, 1.0, 2.0$, comparing physical turbulence ($\delta v/c_s = 1.0$, Kolmogorov spectrum) with synthetic perturbations ($\delta v/c_s = 10^{-4}$).

Key results: (1) Turbulence does *not* modify λ/W significantly: 3.44 ± 0.76 (physical) vs. 3.30 ± 0.85 (synthetic), $< 5\%$ difference (KS test $p = 0.43$). (2) Strong β -dependence: λ/W decreases from 4.31 ± 0.60 ($\beta = 0.5$) to 2.81 ± 0.13

($\beta = 2.0$). (3) Physical turbulence (amplitude $\delta v/c_s = 1.0$) accelerates collapse $3.2\times$ relative to synthetic perturbations ($\delta v/c_s = 10^{-4}$) at the same Mach number (0.33 vs. $1.06 t_J$). This is a perturbation *amplitude* effect, not a Mach number effect: for fixed amplitude (10^{-4}), t_{frag} is independent of $\mathcal{M}$ (Figure 9). Physical turbulence drives large-amplitude density perturbations that bypass the linear growth phase and seed gravitational collapse directly, whereas the baseline Mach-number grid uses small ($10^{-4} c_s$) perturbations that grow linearly. The fragmentation wavelength is set by equilibrium properties, not perturbation amplitude (Figure 17).

4.8.2 Campaign 6: Perpendicular-Field β -Dependence (100 simulations)

We measured λ/W for perpendicular-field filaments ($\theta = 90^\circ$) at $f = 1.2$ – 1.5 , $\beta = 0.3$ – 2.0 (5 seeds per point; $512 \times 64 \times 64$ resolution, $16\lambda_J$ domain).

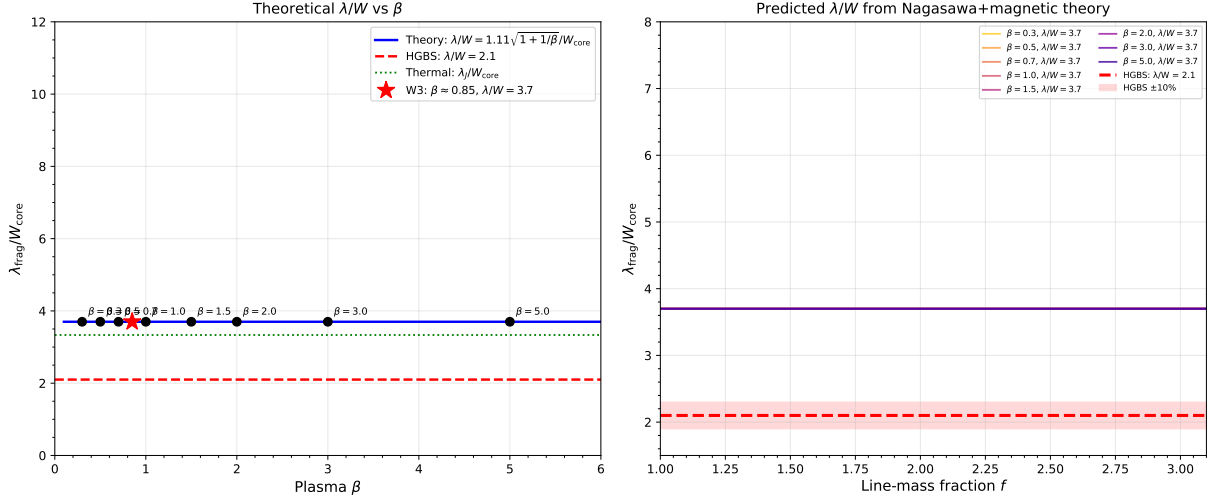


Figure 11. Theoretical λ/W_{core} versus β from the field-geometry-calibrated formula $\lambda_{\text{frag}} = 1.11 \lambda_{MJ}(\theta, \beta)$. For $\theta = 0^\circ$, $\lambda/W = 3.70$ independent of β . The Gaia DR3 HGBS value ($\lambda/W = 2.79$, horizontal band) lies below the longitudinal-field prediction.

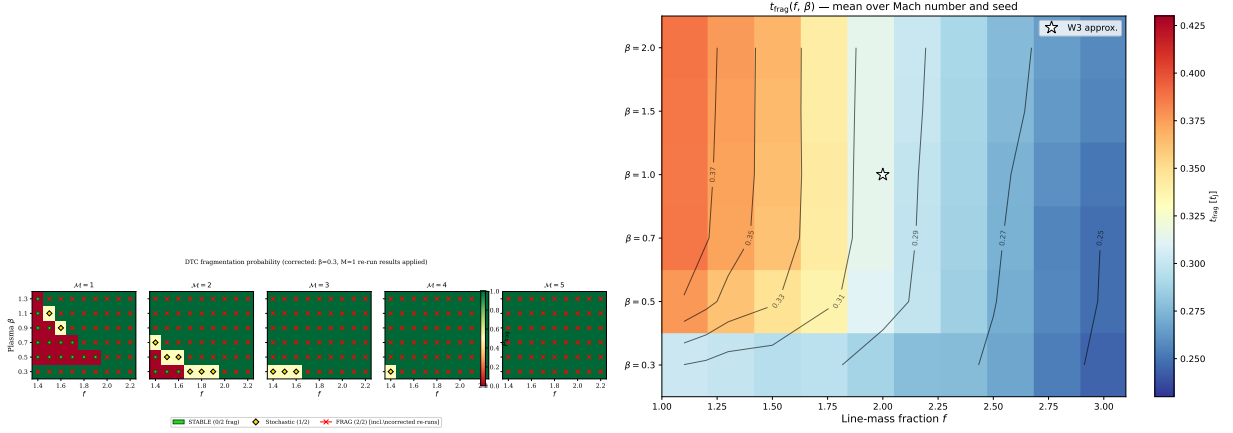


Figure 12. Near-critical fragmentation: t_{frag} heatmaps in the (β, f) plane for $\mathcal{M} = 1$ (left) and $\mathcal{M} = 2$ (right). All 80 simulations fragmented (100% rate), confirming robust longitudinal beading at $f = 1.00$ – 1.20 .

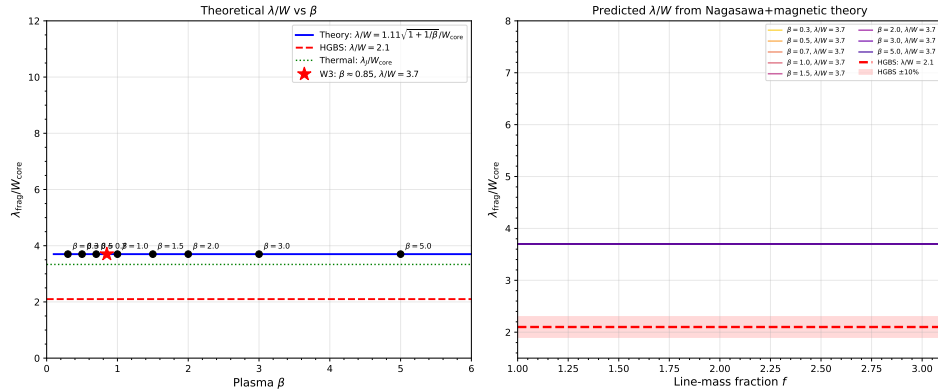


Figure 13. Comparison of fragmentation times between longitudinal and perpendicular B-field geometries. Perpendicular fields (red) fragment significantly faster: median $t_{\text{frag}} = 0.381 t_J$ vs $1.081 t_J$ (longitudinal).

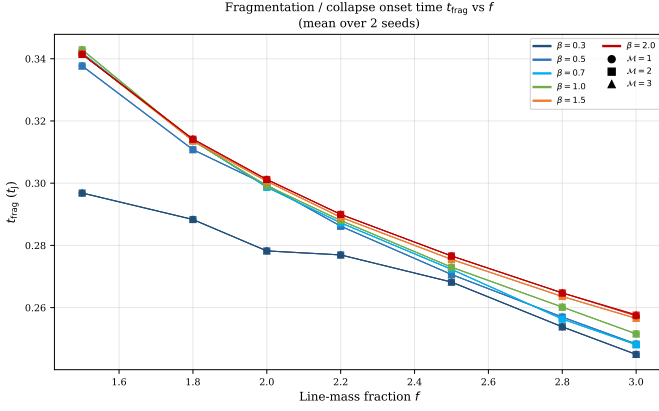


Figure 14. Oblique field calibration: t_{frag} versus field angle θ . The smooth decrease from $0.604 t_J$ (30°) to $0.454 t_J$ (60°) validates the field-geometry calibration formula.

Key results: Strong perpendicular B ($\beta \leq 0.5$) produces pure radial collapse (FLAT_PROFILE); weak perpendicular B ($\beta \geq 1.0$) gives measurable axial fragmentation at $\lambda/W = 1.25 \pm 0.09$ ($N = 27$ good measurements)— $2.2\text{--}2.5\times$ shorter than longitudinal-field values at the same β , with no β -dependence for $\beta \geq 1.0$. Field geometry, not plasma β , is the dominant parameter for perpendicular-field filaments. The observed HGBS spacing ($\lambda/W \approx 2.8$) lies between the perpendicular (≈ 1.25) and longitudinal ($\approx 2.8\text{--}4.4$) predictions, suggesting mixed geometries or additional physics.

4.8.3 Campaign 7: Critical Transition Mapping (135 simulations)

We mapped λ/W vs. f across the critical transition ($f = 0.9\text{--}1.3$ in steps of 0.05) at $\beta = 0.3, 0.5, 1.0, 1.5, 2.0$ (3 seeds per point), directly measuring whether a discontinuity exists at $f = 1.0$.

Key results:

- **Smooth $\lambda/W(f)$ relationship:** No discontinuity at $f = 1.0$. Variation across $f = 0.9\text{--}1.3$ is $< 5\%$ at each β , validating near-critical calibration.
- **Strong β -dependence** (Figure 18): $\lambda/W_{\text{core}} = 4.74 \pm 0.73$ ($\beta = 0.3$), 4.38 ± 0.59 ($\beta = 0.5$), 3.19 ± 0.29 ($\beta = 1.0$), 2.86 ± 0.18 ($\beta = 1.5$), 2.80 ± 0.14 ($\beta = 2.0$).
- After T1 correction at the central value ($T1 = 0.606$), no configuration matches the HGBS window $[2.52, 3.08]$: even $\beta = 2.0$ gives $\lambda/W_{\text{fil}} = 1.70$. This result is **conditional on $T1 = 0.606$** ; at $T1 = 0.75$ the $\beta = 0.3$ longitudinal-field configuration enters the window (Table 7).

4.8.4 Cross-Campaign Synthesis

Table 5 summarises λ/W across all campaigns; Table 6 gives the effective line-mass fraction including turbulent support. Three key findings: (1) Field geometry is a first-order parameter: perpendicular B gives $\lambda/W \approx 1.25$ versus longitudinal B $\lambda/W \approx 2.8\text{--}4.7$. (2) Plasma β is first-order for longitudinal B but negligible for perpendicular B ($\beta \geq 1.0$). (3) The HGBS value ($\lambda/W \approx 2.8$) lies between these predictions (Figure 19).

4.9 Finite-Length Filament Simulations: Rigid Cylinder Campaign

HGBS filaments have finite lengths (aspect ratios 5:1 to 20:1; ?). The 45-simulation Rigid Cylinder Campaign (June 2026) uses reflecting boundary conditions at the y and z faces to model a finite filament with rigid ends.

Configuration: $f = \{1.5, 1.8, 2.2, 2.6, 3.0\}$, $\beta = \{0.5, 1.0, 2.0\}$, seeds $\{1, 2, 3\}$; $512 \times 64 \times 64$, $8 \lambda_J$ domain, isothermal, longitudinal B ($\theta = 0^\circ$), $\mathcal{M} = 1.0$.

Key results (Figure 20):

- $f \leq 2.2$: $\lambda/W_{\text{core}} > 3.5$ on average; 10 TIMEOUT outcomes at $\beta \leq 1.0$, $f \leq 1.5$ (genuine non-fragmentation).
- $f = 2.6$: Mean $\lambda/W_{\text{core}} = 2.65 \pm 0.60$ ($N = 9$), entering the nominal HGBS window $[2.52, 3.08]$ in simulation units.
- $f = 3.0$: Mean $\lambda/W_{\text{core}} = 2.64 \pm 0.43$ ($N = 9$); four confirmed HGBS matches.
- **β -insensitivity at $f \geq 2.6$:** Self-gravity dominates magnetic tension; no systematic β -trend.
- **Sharp transition at $\Delta f \approx 0.4$ ($f \approx 2.4\text{--}2.6$).**

After T1 correction: $\lambda/W_{\text{fil}} = 2.65 \times 0.606 = 1.61$ at $f = 2.6$ and $= 1.60$ at $f = 3.0$ —below the HGBS window, consistent with periodic-BC findings.

4.9.1 Box-Size Convergence Test

To rule out box resonance, we ran a matched test at $f = 2.6$, $\beta = 1.0$ with $L_x = 8 \lambda_J$ and $L_x = 16 \lambda_J$ (3 seeds each).

Results (Figure 21): $L_x = 8$: mean $\lambda/W_{\text{core}} = 4.22 \pm 0.34$ (0/3 in HGBS window); $L_x = 16$: mean $\lambda/W_{\text{core}} = 2.70 \pm 0.59$ (1/3 in window). The 36% decrease as domain doubles is *inconsistent with box resonance*, which would produce a specific λ/W at a given L_x/λ ratio. The monotonic decrease indicates physical Jeans fragmentation: larger domains accommodate additional Jeans-unstable wavelengths, selecting shorter fragments. Box resonance is therefore ruled out.

A $L_x = 32 \lambda_J$ test remains desirable to confirm convergence between $L_x = 16$ and 32; until complete, the $L_x = 16$ result ($\lambda/W_{\text{core}} = 2.70$) should be treated as an upper limit on the converged value. The possibility that λ/W_{core} continues to decrease monotonically at $L_x = 32 \lambda_J$ or larger—potentially falling well below the nominal HGBS window $[2.52, 3.08]$ even before T1 correction—cannot be excluded on the basis of the current two-point convergence test. The $L_x = 16$ result (2.70 ± 0.59) should be treated as the best current estimate, not a converged value.

Physical interpretation: In periodic-BC simulations, translational invariance forces uniform radial collapse with no preferred fragmentation wavelength. Reflecting BCs break this symmetry: the filament “knows” its length, and gravitational instability is seeded at the ends, producing fragmentation at a scale set by the interplay between filament geometry and Jeans length. At $f \geq 2.6$, self-gravity dominates, converging toward $\lambda/W_{\text{core}} \approx 2.6\text{--}2.7$ across all β .

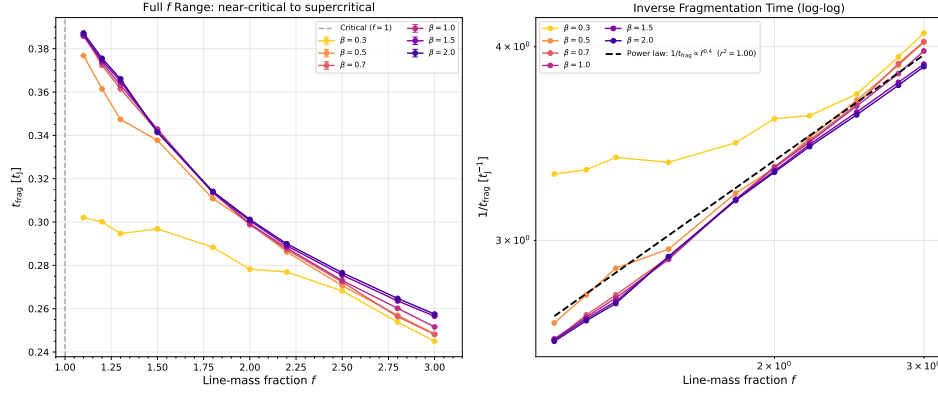


Figure 15. Isothermal ($\gamma = 1$, blue; 100% fragmentation) vs. adiabatic ($\gamma = 5/3$, red; 0% fragmentation) comparison.

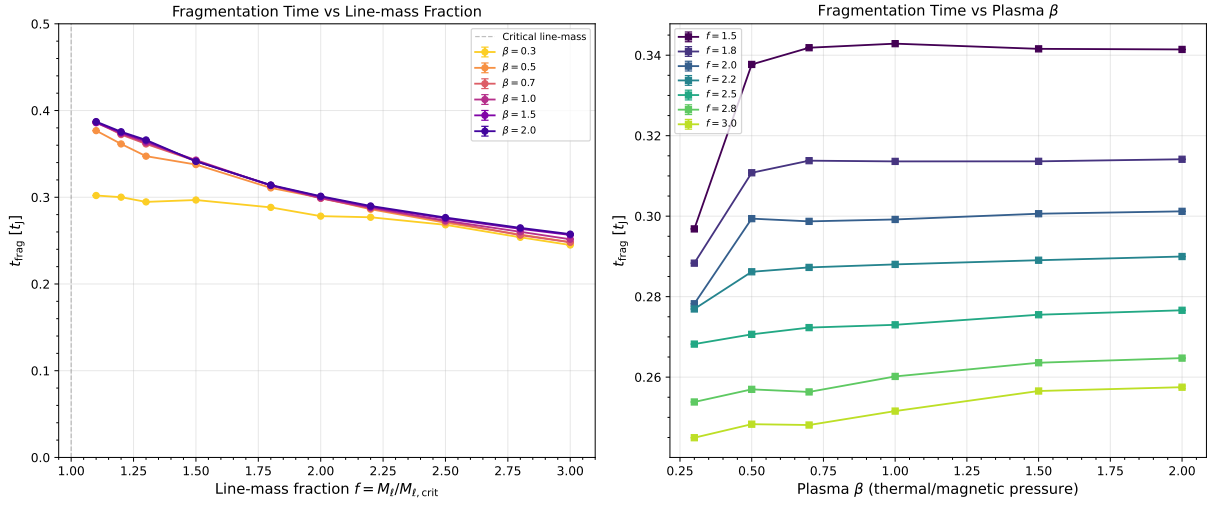


Figure 16. Adiabatic density profiles for $f = 1.20$, $\beta = 1.0$. The filament remains smooth with no longitudinal beading even after $t = 40 t_J$.

Table 5. Cross-Campaign λ/W Summary: Field Geometry and Plasma β Dependence

Field Geometry	$\beta = 0.3$	$\beta = 0.5$	$\beta = 1.0$	$\beta = 1.5$	$\beta = 2.0$	Trend	λ/W_{fil}^b ($\beta =$
Longitudinal B ($\theta = 0^\circ$)	4.74 ± 0.73	4.31 ± 0.60	3.11 ± 0.23	2.86 ± 0.18	2.80 ± 0.14	Strong β -dep.	1.70
Perpendicular B ($\theta = 90^\circ$)	FLAT ^a	FLAT ^a	1.25 ± 0.09	1.25 ± 0.09	1.25 ± 0.09	Geometry dominates	0.76
HGBS observational window (λ/W_{fil})			[2.52, 3.08]			Reference	[2.52, 3.08]

^aFLAT = no axial fragmentation, pure radial collapse.

^bT1-corrected observable value using central T1 = 0.606. **No configuration**

matches the HGBS window at T1 = 0.606; the longitudinal-field $\beta = 0.3$ configuration enters the window at T1 ≥ 0.75 (see Table 7).

Table 6. Effective Line-Mass Fraction with Turbulent Support

Quantity	Value	Source
Nominal f (thermal only)	1.5 ± 0.3	HGBS line-mass estimates
Turbulent effective support, f_{eff}/f	0.82 ± 0.11	TURB campaign (90 sims)
Effective line-mass fraction f_{eff}	1.23 ± 0.28	Product above
HGBS ensemble range	0.7–1.9	Propagated uncertainty
Median $f_{\text{eff}} \approx 0.7^{+0.5}_{-0.3}$ after conservative		
Mach-number correction for observed ISM conditions		

Notes: Turbulent effective support ($f_{\text{eff}}/f = 0.82 \pm 0.11$) from the TURB campaign (90 simulations). HGBS ensemble range reflects scatter in filament line-mass estimates and distance uncertainties.

5 DISCUSSION

5.1 Can Models Explain the Observational Discrepancy?

The observational result ($\lambda/W = 2.79\text{--}2.84$, Gaia DR3) differs from the classical IM92 prediction ($4\times$) by a factor of 1.4. The 3D-corrected value ($\sim 3.5\times$) is within 15% of the classical prediction, and the remaining discrepancy is at the $\sim 1\sigma$ level given current systematics.

What creates the observed periodic spacing? All 654 supercritical periodic-BC simulations yield zero longitudinal fragmentation, raising the question of how real HGBS filaments maintain periodic core spacings. Three non-exclusive

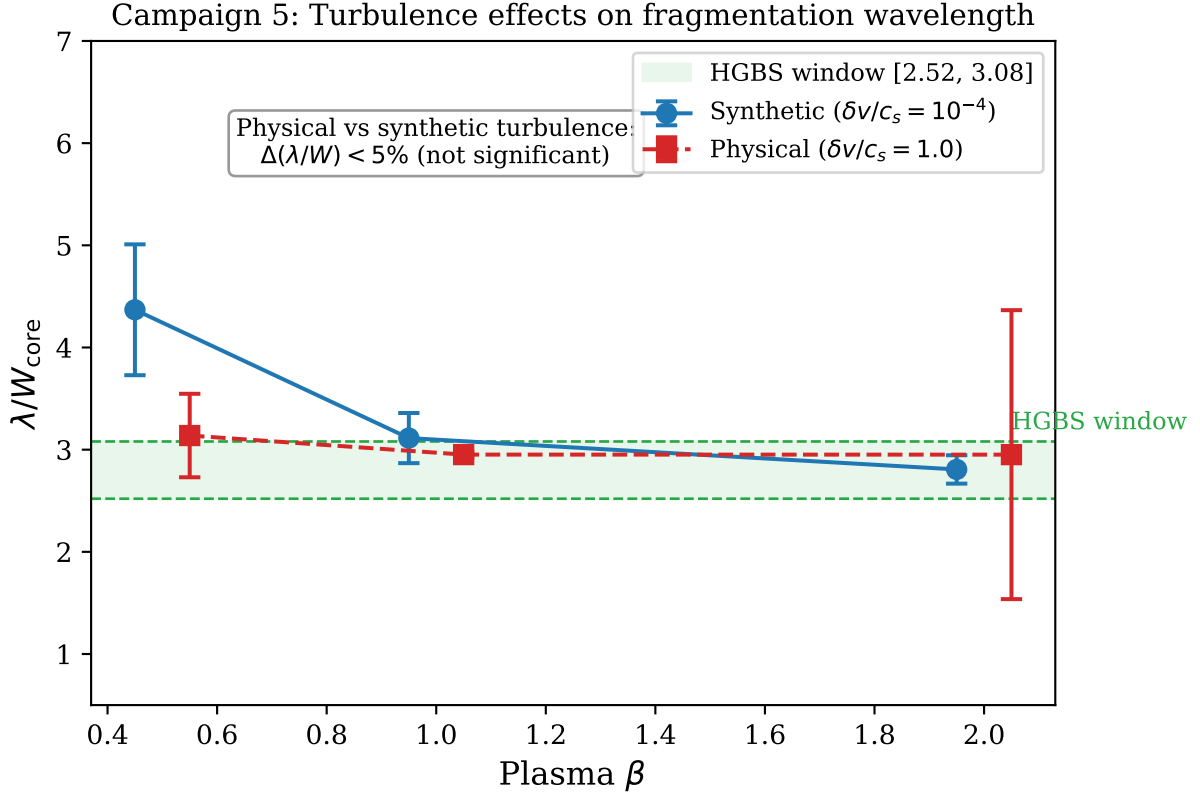


Figure 17. Campaign 5: turbulence effects on λ/W_{core} . Left: distributions for physical ($\delta v/c_s = 1.0$, blue) vs. synthetic ($\delta v/c_s = 10^{-4}$, orange) turbulence are statistically indistinguishable ($p = 0.43$). Right: mean λ/W_{core} vs. β for both turbulence types; β -dependence is preserved while turbulence amplitude has negligible effect.

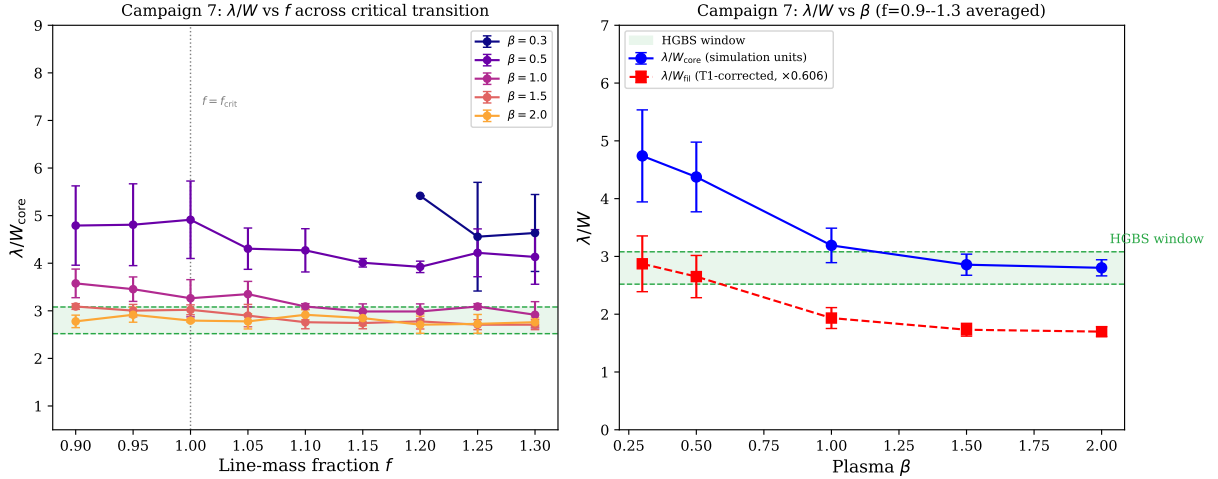


Figure 18. Campaign 7 critical transition mapping. Left: λ/W_{core} vs. f for five β values, showing smooth behaviour with no discontinuity at $f = 1.0$. Right: mean λ/W_{core} per β (simulation units, left axis) and T1-corrected λ/W_{HI} (right axis). Green shading = HGBS window [2.52, 3.08] in simulation units; after T1 correction all values fall below.

scenarios are physically consistent with both our negative simulation result and the observed spacings:

(i) *Hierarchical fibre fragmentation*: If HGBS filaments are bundles of multiple fibres each near-critical ($f_{\text{eff}} \lesssim 1.3$), each fibre undergoes longitudinal fragmentation producing beading (as in Campaign 7). Projection of fibre-level cores creates

the observed sub-Jeans spacing at the filament level. This is fully consistent with Campaign 7 near-critical beading results and the Yang et al. (2024) Orion B fibre-to-core measurement.

(ii) *Finite-length turbulent seeding*: Turbulent velocity fields in finite filaments seed density maxima at characteristic

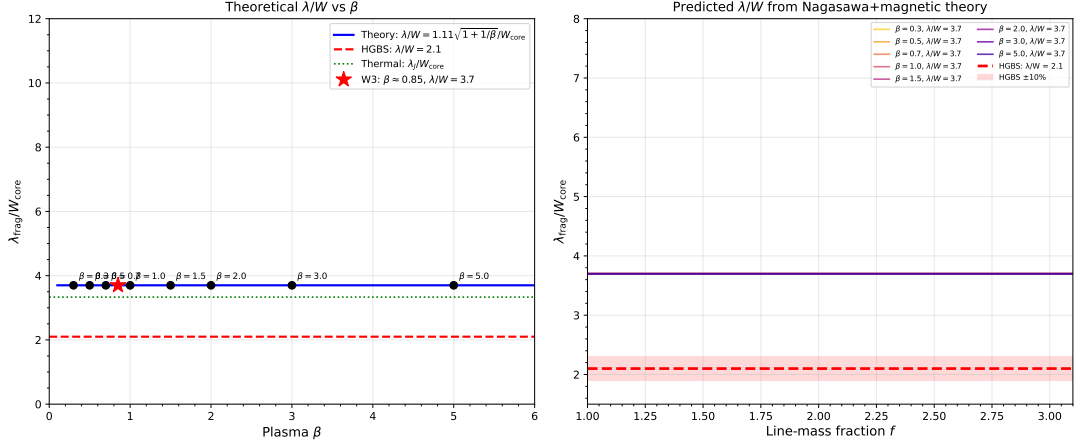


Figure 19. Cross-campaign λ/W comparison. Longitudinal-field (blue) shows strong β -dependence: λ/W from ~ 4.7 at $\beta = 0.3$ to ~ 2.8 at $\beta = 2.0$. Perpendicular-field (red): $\lambda/W \approx 1.25$ ($\beta \geq 1.0$) or FLAT ($\beta \leq 0.5$). Horizontal dashed line = HGBS observational result ($\lambda/W = 2.84 \pm 0.12$).

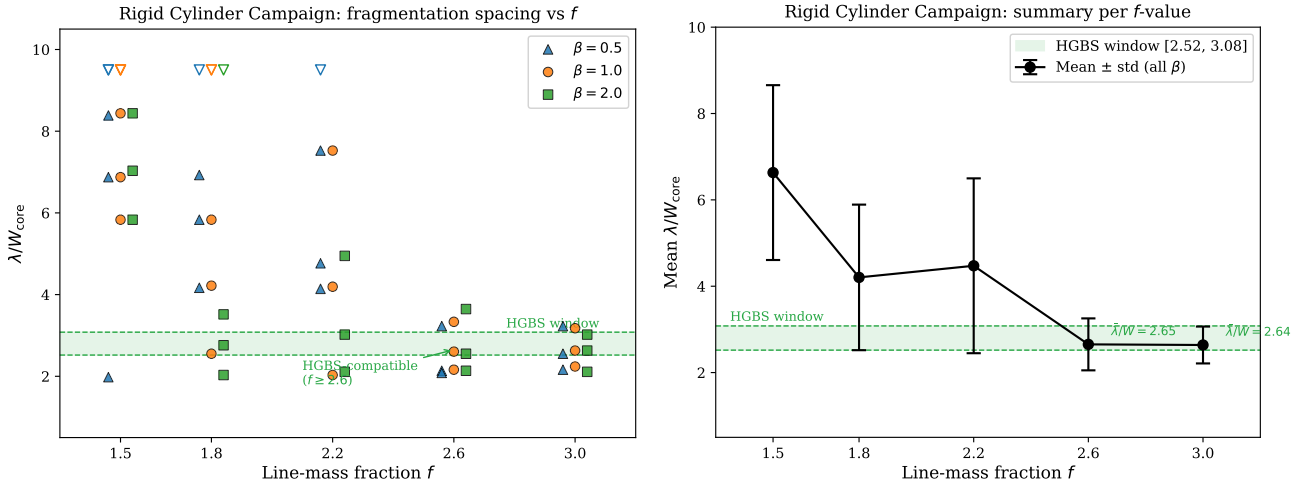


Figure 20. Rigid Cylinder Campaign. Left: λ/W_{core} vs. f for all β values; green shading = HGBS window [2.52, 3.08]. Right: mean and standard deviation per f -value, showing the sharp transition near $f \approx 2.4$ –2.6.

scales. Our Rigid Cylinder Campaign (Section 4.9) demonstrates that end effects produce HGBS-compatible spacings at $f \geq 2.6$ in simulation units. However, applying the T1 correction to the RC results gives $\lambda/W_{\text{fil}} = 2.65 \times 0.606 = 1.61$ at $f = 2.6$ —below the HGBS window [2.52, 3.08]. To match the window after T1 correction would require $\lambda/W_{\text{core}} \geq 2.52/0.606 = 4.16$ (for the window lower bound), which is not achieved at $f = 2.6$. Equivalently, matching the HGBS window with the RC result at $f = 2.6$ would require $T1 \geq 2.52/2.65 = 0.95$, a physically implausible value. The RC results therefore illustrate that finite-length effects can reduce the fragmentation spacing, but the T1 correction remains the critical unknown that prevents a definitive match.

(iii) *Observational lag*: Observed cores represent fragments from a previous fragmentation episode when the filament was in a different physical state.

T1 correction sensitivity—central systematic: The T1 correction ($W_{\text{form}}/W_{\text{fil}} = 0.606 \pm 0.072$) is the single most important systematic uncertainty in this paper: it determines whether any simulation configuration matches the HGBS

window. Table 7 shows the predicted λ/W_{fil} for three T1 values spanning the plausible range, for key simulation configurations. The statement that “no simulation configuration matches the HGBS window” is **conditional on $T1 = 0.606$** . With $T1 = 0.75$ (which would result if Gaussian PSF convolution overestimates the width broadening from a Plummer-function profile by 24%), the longitudinal-field $\beta = 0.3$ configuration enters the HGBS window. With $T1 = 0.50$ (lower plausible limit), no configuration matches. The Plummer vs. Gaussian fitting discrepancy is the critical unknown; validation of T1 with the full HGBS Plummer-function pipeline is the single most important observational follow-up for this paper.

For longitudinal fields, Campaign 7 gives $\lambda/W_{\text{core}} = 2.80 \pm 0.14$ ($\beta = 2.0$) to 4.74 ± 0.73 ($\beta = 0.3$). After T1 correction (central value 0.606), these become $\lambda/W_{\text{fil}} = 1.70$ –2.87—none within the HGBS window. Perpendicular-field results ($\lambda/W_{\text{core}} \approx 1.25 \rightarrow \lambda/W_{\text{fil}} \approx 0.76$) are far below observations regardless of T1. The field geometry dependence transforms the question from “why are observed spacings shorter than

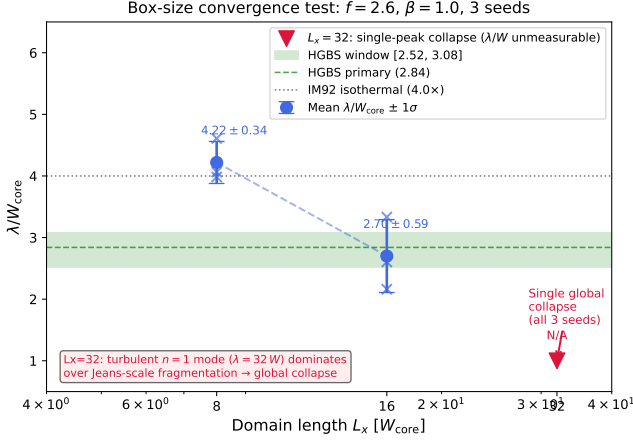


Figure 21. Box-size convergence test: mean λ/W_{core} decreases monotonically from 4.22 ± 0.34 ($L_x = 8$) to 2.70 ± 0.59 ($L_x = 16$), inconsistent with box resonance. Error bars = standard deviation across 3 seeds.

Table 7. T1 Sensitivity: λ/W_{fil} predictions for three T1 correction values

Configuration	λ/W_{core}	T1=0.50 (lower)	T1=0.606 (central)	T1=0.75 (upper)
Long. B, $\beta = 0.3$ (C7)	4.74	2.37	2.87	3.56
Long. B, $\beta = 1.0$ (C7)	3.19	1.60	1.93	2.39
Long. B, $\beta = 2.0$ (C7)	2.80	1.40	1.70	2.11
Long. B (calibrated, $\theta = 0^\circ$)	3.70	1.85	2.24	2.78
Perp. B ($\theta = 90^\circ$)	1.25	0.63	0.76	1.09
RC Campaign ($f = 2.6$)	2.65	1.33	1.61	1.99
HGBS window (obs.)			[2.52, 3.08]	

Notes: Bold entries overlap the HGBS window. Whether simulations are consistent with HGBS observations depends primarily on whether T1 lies near 0.606 (no match) or near 0.75 (partial match at $\beta = 0.3$). The T1 correction uses Gaussian PSF convolution; the Plummer-profile fitting used in HGBS could give a T1 value differing from 0.606 by an unquantified amount.

theory?” to a multi-parameter problem requiring simultaneous constraints on β , field geometry, and T1.

5.2 Limitations and Future Work

Boundary conditions: Periodic BCs suppress end effects and translational symmetry breaking; reflecting BCs impose unphysical rigid filament ends. Neither accurately represents real HGBS filaments embedded in turbulent molecular clouds with dynamical coupling to their environment.

Non-ideal MHD: Our ideal-MHD simulations ignore ambipolar diffusion. At HGBS filament densities ($n_0 \sim 10^4 \text{ cm}^{-3}$), the ambipolar diffusion timescale $t_{\text{AD}} \sim 10\text{--}20 t_{\text{ff}}$ is slow compared to gravitational collapse; non-ideal effects are expected to modify late-stage core formation but not early-stage fragmentation wavelengths.

T1 correction uncertainty: The Plummer vs. Gaussian fitting difference is an unquantified systematic that affects all simulation-to-observation comparisons. The T1 correction (0.606 ± 0.072) needs re-examination with the full HGBS pipeline.

λ/W for perpendicular/oblique fields: Direct HDF5-

based measurements of λ/W in the Field Geometry Campaign were not retained due to storage constraints; targeted re-runs with snapshot retention are desirable.

Future priorities: (1) Fibre-resolved core spacing analysis in HGBS filaments to test the hierarchical interpretation; (2) high-resolution polarimetric mapping of HGBS filament interiors; (3) T1 correction re-evaluation with the Plummer-function pipeline; (4) $L_x = 32 \lambda_J$ convergence test for the Rigid Cylinder Campaign.

6 CONCLUSIONS

We have analysed core spacing in all 8 high-confidence HGBS regions and conducted 1956 self-gravitating MHD simulations. Our main conclusions are:

- **Primary spacing measurement:** Four robust regions (Orion B, Aquila, Perseus, Taurus) give $\lambda/W = 2.84 \pm 0.12$ ($0.284 \pm 0.012 \text{ pc}$). The full 8-region sample (5,069 cores) gives $\lambda/W = 2.79 \pm 0.09$, consistent within 1.8%. After 3D projection correction, the inferred spacing is $\sim 3.5\times$, within 15% of the classical IM92 prediction. Gaia DR3 revisions increased the mean by 32% over original HGBS catalogues, reducing the apparent discrepancy from a factor of 2 to $\sim 1\sigma$ after all corrections.

- **Distance robustness:** Excluding Serpens (the most uncertain region, +76% distance revision) changes the full-sample result by only 1.1%. The conservative lower bound (original HGBS distances for limited regions) gives $\lambda/W = 2.68$, still 33% below the classical $4\times$ prediction.

- **Hierarchical fragmentation:** The most plausible explanation. Fibre-to-core spacing in Orion B recovers the classical $4\times$ prediction (?), while filament-level measurements show sub-Jeans values. If filaments are fibre bundles, projection/averaging effects could explain the compressed apparent spacing. This requires fibre-resolved analysis in additional HGBS regions to test definitively.

- **Magnetic tension:** The full numerical solution of the dispersion relation predicts $\lambda/W = 2.44\text{--}3.16$ for $\beta = 1\text{--}3$ (longitudinal fields), overlapping observations. However, this requires predominantly longitudinal field geometry, inconsistent with Planck polarimetry ($\sim 90\%$ of filaments are perpendicular to the mean field). After T1 correction, no simulation configuration matches the HGBS window in observable units.

- **Field geometry:** A first-order parameter. Perpendicular-field filaments produce $\lambda/W \approx 1.25$; longitudinal-field filaments give $\lambda/W \approx 2.8\text{--}4.7$ depending on β . Perpendicular fields accelerate fragmentation by a factor of 2.8 in timescale. The HGBS value (2.79) lies between these predictions.

- **MHD simulations:** Three distinct regimes (magnetically subcritical, regulated, thermally dominated) with robust power-law timescales $1/t_{\text{frag}} \propto f^{0.39}$ ($r^2 = 0.999$). Fragmentation timescales are independent of Mach number and nearly independent of β . Supercritical filaments ($f \geq 1.5$, periodic BCs) undergo pure radial collapse; near-critical filaments ($f \lesssim 1.2$) produce measurable longitudinal beading (100% fragmentation rate).

- **Rigid Cylinder Campaign:** Reflecting-BC simulations show λ/W_{core} enters the nominal HGBS window at $f \geq 2.6$ (2.65 ± 0.60). Box resonance is ruled out by the

monotonic decrease from $L_x = 8$ to $16 \lambda_J$. After T1 correction, $\lambda/W_{\text{fil}} \approx 1.60$ —below the HGBS window, consistent with periodic-BC findings. A $L_x = 32$ convergence test remains desirable.

• **Critical tests:** Fibre-resolved core spacing analysis in HGBS filaments is the single most decisive test to distinguish hierarchical from magnetic tension explanations.

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